

4.3 TYPES OF COMMUNICATION SYSTEMS

- One may categorize communication systems based on their physical infrastructure and the specifications of the signals they transmit.
- The physical infrastructure pertains to the type of the channel used and the hardware design of the transmitting and receiving equipment.
- The signal specifications signify the nature and type of the transmitted signal.
- **Based on the physical infrastructure**, there are two types of communication systems:
 - (i) In **Line Communication Systems**, there is a physical link, called the hardware channel, between the transmitter and the receiver in the line communication systems.
Eg: landline telephony is purely a line communication system
 - (ii) In a **radio communication system**, there is no such link and natural resources, such as space and water are used as software channels. e.g. radio broadcast is a purely radio communication system

- **Communication Systems Based on Signal Specifications**

The signal specifications used to decide the type of communication include:

- a) Nature of baseband or informal signal
- b) Nature of the transmitted signal.
 - **Based on the nature of the baseband signal**, there are two types of communication systems:
 - **Analog communication systems**
 - **Digital communication systems.**
 - **Based on the nature of the transmitted signal**, the baseband signal can either be transmitted as it is, without modulation, or through a carrier signal with modulation. The two systems can then be put under following categories:
 - **Baseband communication system** – signal transmitted without modulation
 - **Carrier communication system** – signal transmitted with modulation using carrier signal.

Thus, there are four types of communication system categories based on signal specification. These are:

- Analog communication system eg: TV transmission
- Digital communication system eg: high definition television (HDTV)
- Baseband communication system eg: landline telephony and Fax
- Carrier communication system eg: TV transmission, radio broadcast, and cable TV.

4.4 MODULATION

➤ AMPLITUDE MODULATION

- **Amplitude modulation is defined as the modulation technique in which the instantaneous amplitude of the carrier signal is varied in accordance with the instantaneous amplitude of the analog modulating signal to be transmitted, thus the baseband signal is translated to a higher frequency spectrum.**

- The modulating signal is the analog baseband signal which is random and has low frequency while the carrier signal is always a sinusoidal wave with high frequency.

- The variations in the amplitude of carrier signal represent the information carried.

- Figure 2.1 shows the high frequency carrier signal, modulating signal and the modulated signal. It can be seen from the figure that the amplitude of the carrier signal is varies in accordance with the modulating signal while the frequency and the phase of the carrier signal remain unchanged.

- The modulating signal seems to be superimposed on the carrier signal.

- The amplitude variations in the peak values of the carrier signal exactly replicate the modulating signal at different points in time which is known as an envelope.

- Amplitude modulation has various applications which include radio broadcasting, television broadcasting, satellite communication, etc

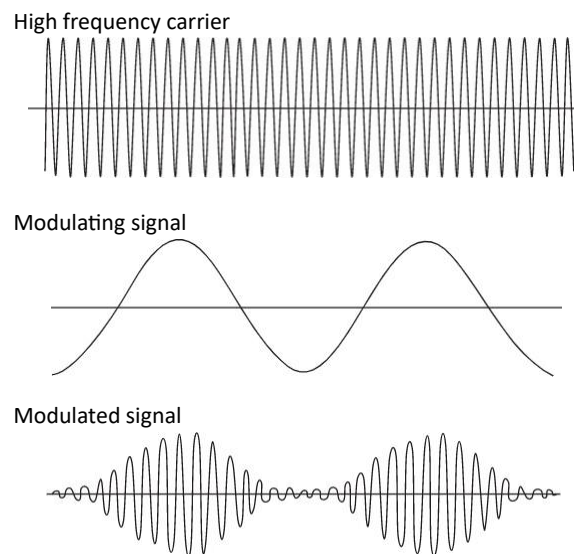
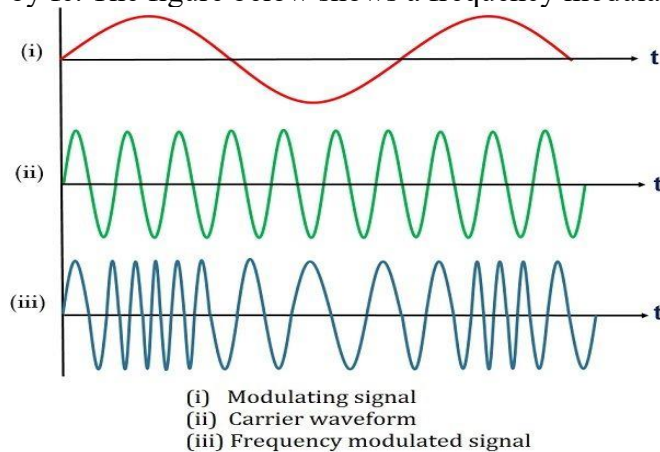


Fig. 2.1 Waveforms related to amplitude modulation

➤ FREQUENCY MODULATION

- A modulating signal may vary the frequency of the carrier keeping the amplitude and phase constant. This type of modulation is called frequency modulation. i.e; the frequency modulation is the process of changing the frequency of the carrier voltage in accordance with the instantaneous value of the modulating voltage.
- The original frequency of the carrier signal is called centre or resting frequency and denoted by f_c . The figure below shows a frequency modulated signal.

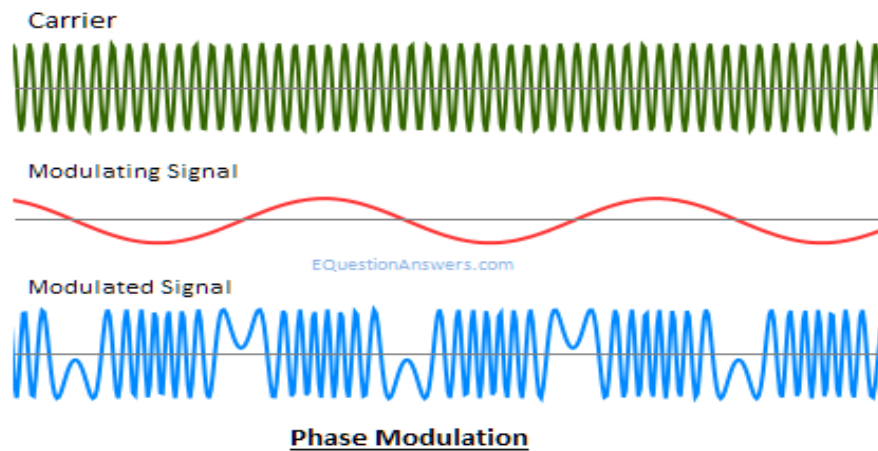


Electronics Coach

- The amount by which the frequency of the carrier wave changes or shifts above or below the resting frequency is termed as **frequency deviation**.

➤ PHASE MODULATION

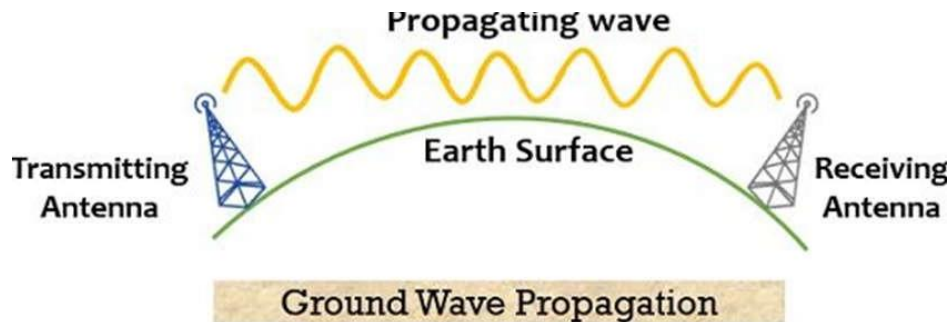
- Phase modulation and frequency modulation are closely related to each other. In both the cases, the total phase angle (t) of the modulated signal varies.
- **Phase modulation is the process in which the instantaneous phase of the carrier signal is varied in accordance with the instantaneous amplitude of the modulating signal. In this type of modulation, the amplitude and frequency of the carrier signal remains constant.**
- The modulating signal is mapped to the carrier signal in the form of variations in the instantaneous phase of the carrier signal.
- The figure below shows a frequency modulated signal.



4.5 RADIO WAVE PROPAGATION

- In wireless communication electromagnetic waves of different frequencies are used to carry information through the physical space acting as the transmission medium. Electromagnetic waves with frequencies extending from about 3 kHz to 300 GHz are classified as radio waves.
- Radio waves are a type of electromagnetic radiation with the longest wavelengths in the electromagnetic spectrum.
- Like all electromagnetic waves, radio waves in vacuum travel at the speed of light (3×10^8 m/s), and in the Earth's atmosphere at a close, but slightly lower speed.
- Radio waves are generated by charged particles undergoing acceleration, such as time-varying electric currents.
- Depending primarily on the frequency a radio wave travels from the transmitting to the receiving antenna in several ways.
- On the basis of the mode of propagation, radio waves can be broadly classified as:
 - (i) ground or surface waves (ii) space or tropospheric waves, and (iii) sky waves.
 Accordingly, we have three types of propagation (modes of propagation of electromagnetic waves):
 - 1) **Ground wave propagation**
 - 2) **Sky wave propagation**
 - 3) **Space wave propagation**

➤ **Ground Wave (or Surface Wave) Propagation:**

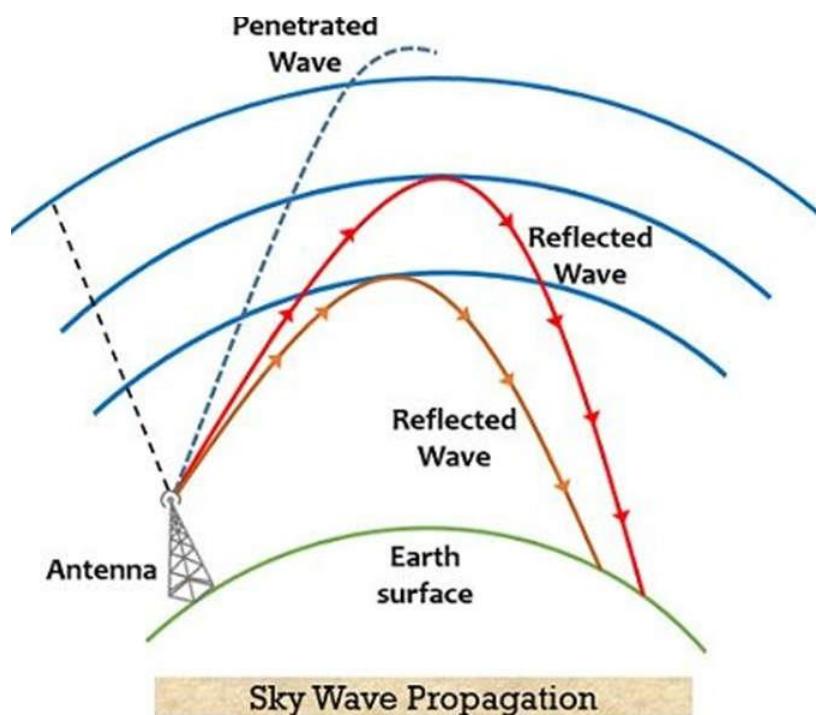


- In ground wave propagation, radio waves are guided along the curved surface of the earth from the transmitter to the receiver.
- As the waves move over the ground, they are strongly influenced by the electrical properties of the ground.
- As high frequency waves are strongly absorbed by ground, ground wave propagation is useful only at low and medium frequencies. So Ground wave propagation is also known as **medium wave propagation**. (usually used for frequencies up to 2MHz)
- Below 500 kHz, ground waves can be used for communication within distances of about 1500 km from the transmitter.
- AM radio broadcast in the medium frequency band cover local areas and take place primarily by the ground wave.
- The ground waves at higher frequencies employed by frequency modulation (FM) and television (TV) are increasingly absorbed and therefore become very weak beyond a distance of several kilometres from the transmitter.
- Ground wave transmission is very reliable whatever the atmospheric conditions be.
- Application: Short distance communications like local radio broadcasting, radar, amateur radio communications, radio navigation etc. employ ground wave propagation

➤ **Sky Wave (or Ionospheric) Propagation:**

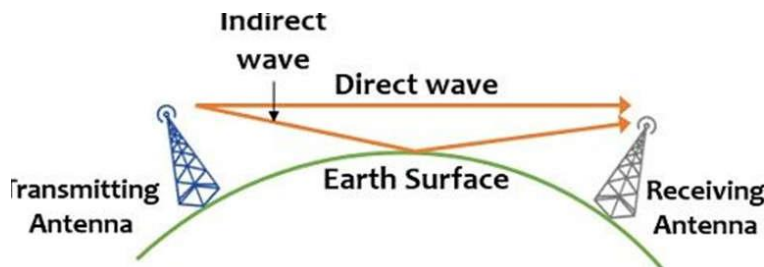
- In this mode of propagation, radio waves transmitted from the transmitting antenna reach the receiving antenna after reflection from the ionosphere, i.e. the ionized layers lying in the earth's upper atmosphere. Short wave transmission around the globe is possible through sky wave via successive reflections at the ionosphere and the earth's surface.

- Fig. below depicts how a radio wave is directed towards the sky and reflected by the ionosphere toward the desired destination of the earth.
- The ionized region of the earth's upper atmosphere extending from about 40 km to the height of a few earth radii above the earth, is referred to as the **ionosphere**. The ionosphere is made up of electrons, and positive and negative ions in the background of neutral particles of the atmosphere.
- The propagation of radio wave through the ionosphere is affected by the electrons and ions in the ionosphere. The effect of the electrons on the propagation is much greater than that of the ions since the electronic mass is much less than the ionic mass.
- Sky wave propagation is mainly used in HF bands (usually from 2MHz to 30MHz)
- Very high frequency waves penetrate through the ionospheric layers and are not reflected back. So Sky wave propagation is suitable for high frequency (short wave) propagation.
- Applications: Used in Mobile Networks, Satellite Networks



➤ Space wave (or tropospheric or Line of Sight) propagation:

- When a radio wave transmitted from an antenna, travelling in a straight line directly reaches the receiving antenna, it is termed as space or tropospheric wave.
- In space wave or line of sight propagation, radio waves move in the earth's troposphere within about 15 km over the surface of the earth.
- The space wave is made up of two components: (a) a direct or Line-of Sight(LOS) wave form the transmitting to the receiving antenna and (b) the ground-reflected wave traversing from the transmitting antenna to ground and reflected to the receiving antenna.
- Space wave propagation is mainly used in VHF bands (between 30Mz and 300MHz).Also used in UHF and microwave frequencies.
- Television frequencies in the range 100-220 MHz are transmitted through this mode.
- Applications: Communications using Satellite, Communication using Radar, Microwave Propagation, TV broadcasting etc.



- **Space Wave Propagation**

4.5 SATELLITE COMMUNICATION

- A satellite is a body that orbits around another body in space. There are two different types of satellites – natural satellites (eg; moon) and man-made or artificial satellite (eg: INSAT 1B, GSAT-31 etc).
- **Satellites are classified into different types based on their applications.**

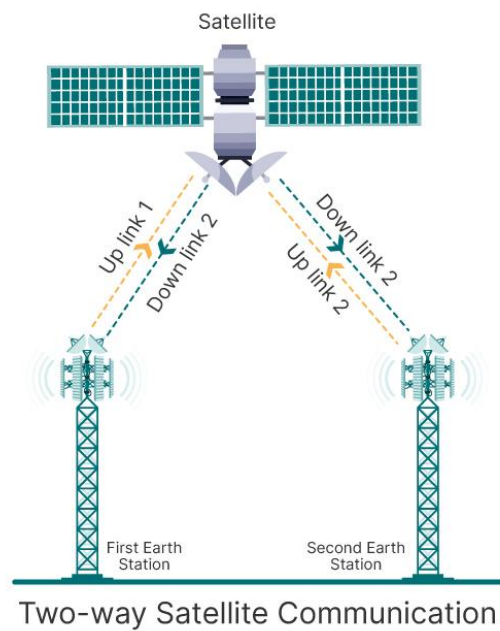
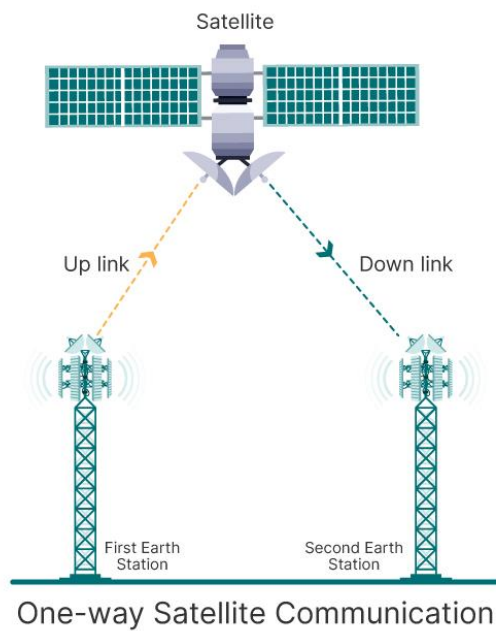
Some of them are:

- i. **Weather Satellites:** They are used to monitor the weather and climate of Earth. By measuring cloud mass, these satellites enable us to predict rain and dangerous storms like hurricanes, cyclones etc.
- ii **Communication satellites:** They are used to transmit television, radio, internet signals etc. Multiple satellites are used for long distances.

- iii. **Navigation satellites:** These are employed to determine the geographic location of ships, aircrafts or any other object.
- ***Satellite communications*** is the process by which radio frequency (RF) signals are sent to and from satellites orbiting the Earth, to enable communication between distant points on the globe.
- The satellite communication is a mode of communication of signal between transmitter and receiver via satellite. The message signal from the Earth station is transmitted to the satellite on board via an uplink (frequency band 6 GHz), amplified by a transponder and then retransmitted to another Earth station via a downlink (frequency band 4 GHz)
- The high frequency radio waves used for communication links travel by line of sight, and so are obstructed by the curvature of the Earth.
- The purpose of communication satellites is to relay and amplify these signal, using a transponder around the curve of the Earth enabling communication between widely separated geographical points.

➤ **Types of Satellite Communication**

- Satellite communications can be broadly categorized into two types based on the direction of communication: **one-way and two-way satellite communication**. As the name implies, “one-way” transmits signals in only one direction –between the satellite and the ground station. This communication method is widely used for radio and television signals, where the satellite simultaneously sends signals to numerous ground stations. Direct broadcast satellite (DBS) and satellite radio are prime examples of one-way satellite communication.
- In contrast, “two-way” is a communication method in which signals are transmitted in both directions – between the satellite and the ground station. This method is employed in various applications, including commercial and military communications, internet access, and remote monitoring.



- The access to remote areas not covered by conventional cable or fiber communications is also a distinct feature of satellites.
- The basic structure of a satellite communication system is shown in Fig. 24.7.

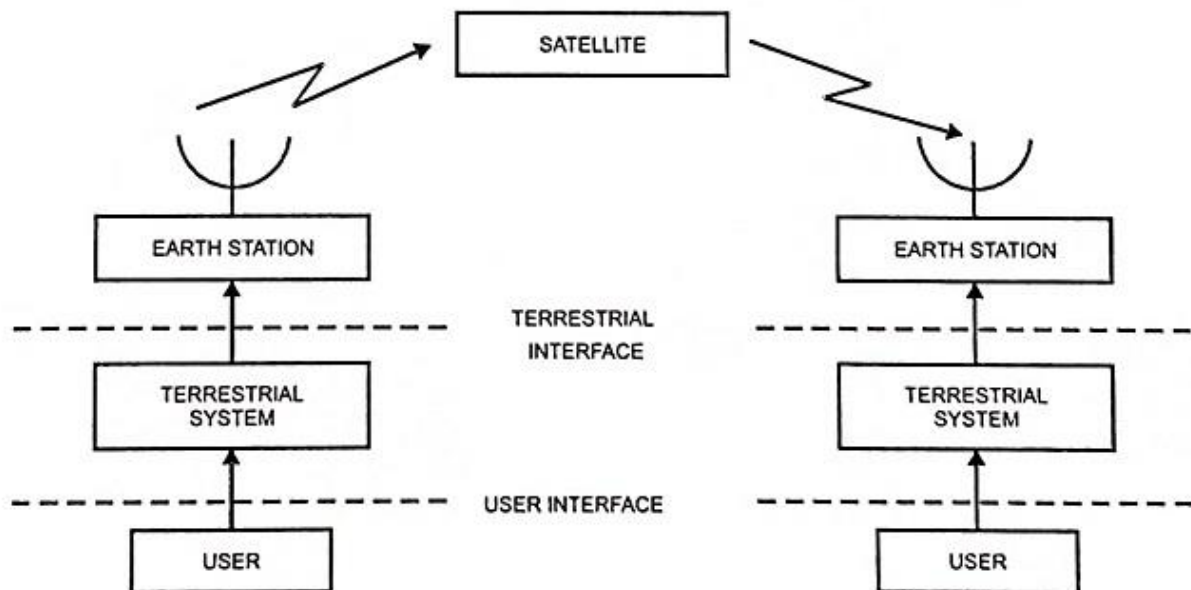


Fig. 24.7 *Basic Structure of a Satellite Communication System*

- Satellite communication system consist of many earth stations on the ground and these are linked with a satellite in space.
- The user generates a baseband signal that is processed through a terrestrial network and transmitted to a satellite at the earth station.
- The satellite consists of a large number of repeaters in space, receives the modulated RF carrier in its uplink frequency spectrum from all the earth stations in the network, amplifies these carriers and retransmits them back to the earth stations in the downlink frequency spectrum.
- To avoid the interference downlink frequency spectrum should be different from uplink frequency spectrum. The signal at the receiving earth stations is processed to get back the baseband signal, it is sent to the user through a terrestrial network.
- Commercial satellite communication system use a frequency band of 500 MHz bandwidth near 6 GHz for uplink transmission and another 500 MHz bandwidth near 4 GHz for downlink transmission. An uplink of 5.725 to 7.075 GHz and a downlink of 3.4 to 4.8 GHz is used
- 14/12 GHz band is also used in commercial communication satellites because of over crowding of geostationary satellites at 6/4 GHz band. Here uplink is of 12.75 to 14.8 GHz and downlink of 10.7 to 12.3 GHz. This band may be extensively used in future.
- A third band is the 30/20 GHz band where 2.5 GHz bandwidth has been allocated with a uplink of 27.5 to 31 GHz and downlink of 18.1 to 21.2 GHz.

➤ **Reasons for the Downlink Frequency to be Lower than the Uplink Frequency**

Frequency of downlink is smaller than that of the uplink, several factors are there such as

- **Output power amplifier in transponder** : It is the most important factor because the final power amplifier in the transponder generates more power at lower frequencies than at higher frequencies.
- **Effective area of the receiving antenna** : The effective capture area of the downlink antenna should be more so as to receive more energy at the earth's receiving antenna. Effective area of an antenna is directly proportional to the square of the frequency. Thus, the frequency should be kept lower.
- **Path loss** : It is less at lower frequencies than at higher frequencies.
- **Beamwidth** : It should be wider as a satellite would send energy to a large number of earth stations. Lower the frequency wider is the beamwidth.

- The basic block diagram of an earth station transmitter is shown in Fig. 24.8.

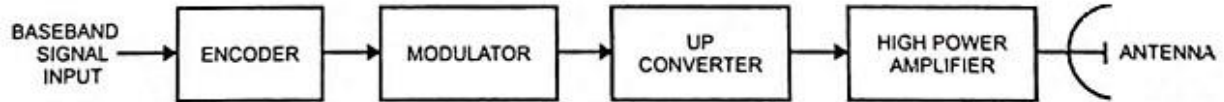


Fig. 24.8 Block Diagram of an Earth Station Transmitter

- The baseband signal from the terrestrial network is processed through encoder and modulator, then it is converted to uplink frequency. Finally it is amplified by high power amplifier and directed towards the antenna.
- The block diagram of an earth station receiver is shown in Fig. 24.9.

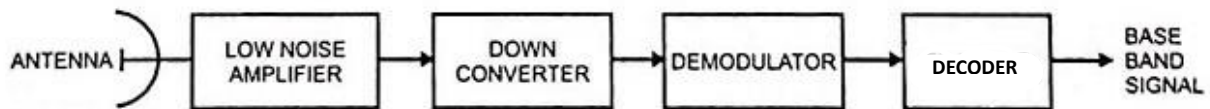


Fig. 24.9 Block Diagram of an Earth Station Receiver

- The signal received from the satellite is processed through low noise amplifier and then it is down converted, demodulated by demodulator and decoded by decoder. Thus the original baseband signal is obtained.

➤ Advantages of Satellite Communication System:

- **Scalability:** Satellite networks can be easily expanded or contracted depending on the user's needs.
- **Global coverage:** This technology can cover any part of the world, including remote and inaccessible areas.
- **Cost-effective:** This technology can be more cost-effective than laying cables or building terrestrial networks.
- **Mobility:** It allows for communication while in motion, making it useful for transportation, aviation, and maritime industries.
- **High-speed connectivity:** This technology provides high-speed data and voice communication services, even in remote areas.
- **Security:** This type of communication is more secure than other forms of communication since the signals are difficult to intercept.
- **Reliability:** It is not affected by terrestrial factors such as terrain, weather, or distance, making it more reliable than other forms of communication.

- **Disaster recovery:** During natural disasters or emergencies, this technology can act as a backup communication system when terrestrial networks are damaged or unavailable.

➤ **Disadvantages of Satellite Communication**

- **Cost:** Setting up and maintaining this technology can be expensive, which makes it less accessible for some organizations.
- **Delay:** The time taken by a signal to travel to and from a satellite can cause delays, which is not favourable in certain situations.
- **Weather:** Adverse weather conditions, such as heavy rain or snow, can disrupt or completely interrupt satellite signals, thus affecting quality and reliability.
- **Vulnerability:** Satellites are vulnerable to space debris, solar flares, and other potential hazards that disrupt communication, making them less reliable in certain scenarios.

➤ **Applications of Satellite Communication**

- **Navigation:** Satellites help in navigation and location tracking, such as air and sea transportation, surveying, military operations, etc.
- **Internet access:** Satellites provide internet access in remote or rural areas where other forms of connectivity are unavailable or unreliable.
- **Broadcasting:** Satellites distribute TV and radio signals, allowing broadcasters to reach a wider audience beyond terrestrial transmission capabilities.
- **Military and defence:** Satellites are used for various military and defence applications, such as communication, surveillance, intelligence gathering, etc.
- **Telecommunications:** This type of communication allows long-distance voice and data communication in remote or isolated regions where another communication infrastructure is unavailable.
- **Remote sensing:** Satellites equipped with sensors can collect data about the Earth's surface, atmosphere, oceans, etc. And this information is used for various purposes, such as weather forecasting, natural disaster monitoring, agricultural monitoring, etc.

4.6 MOBILE COMMUNICATION

- **Mobile Communication** is the use of technology that allows us to communicate with others in different locations without the use of any physical connection (wires or cables). Mobile communication makes our life easier, and it saves time and effort.
- A **mobile phone** (cell phone or hand phone) is an example of mobile communication (wireless communication). It is an electric device used for full duplex two way radio telecommunication over a cellular network of base stations known as cell site.
- **Mobile communication** system may be defined as a communication system that allows people to communicate without utilizing any physical link, disregarding, location, time, and distance.
- **GSM (or Global System for Mobile Communications)** is defined as a set of mobile communications standards and protocols governing second-generation or 2G networks

➤ GSM network architecture

- The GSM network architecture can be grouped into four main areas:
- **Network and Switching Subsystem (NSS)**
- **Base-Station Subsystem (BSS)**
- **Mobile station (MS)**
- **Operation and Support Subsystem (OSS)**

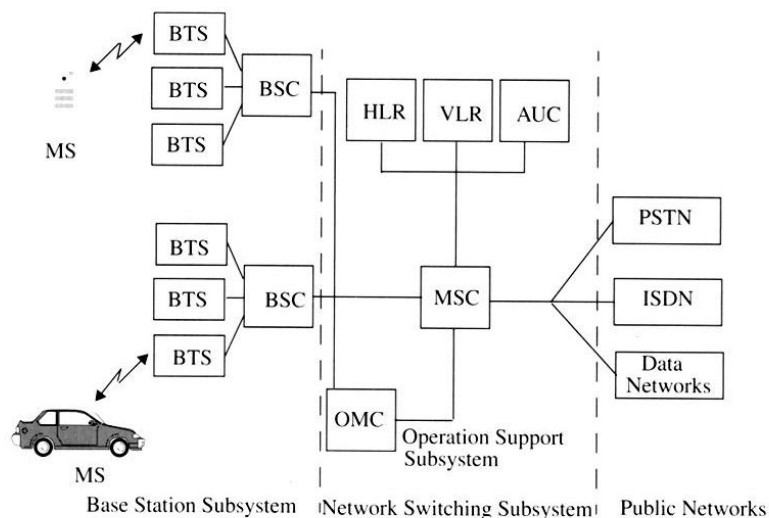


Figure 11.5 GSM system architecture.

- The different elements of the GSM network operate together and the user is not aware of the different entities within the system.
- **Network Switching Subsystem (NSS)**
- The GSM system architecture contains a variety of different elements, and NSS is often termed the core network. It is essentially a data network with a various entities that provide the main control and interfacing for the whole mobile network. The major elements within the core network include:
 - **Mobile Services Switching Centre (MSC):**
 - The main element within the core network area of the overall GSM network architecture is the Mobile switching Services Centre (MSC).
 - The MSC acts like a normal switching node within a PSTN or ISDN, but also provides additional functionality to enable the requirements of a mobile user to be supported. These include **registration, authentication, call location, inter-MSC handovers and call routing to a mobile subscriber.**
 - It also provides an interface to the PSTN so that the **mobile communications calls can be routed from the mobile network to a phone connected to a landline.** Interfaces to other MSCs are provided to **enable calls to be made to mobiles on different networks.**
 - **Home Location Register (HLR): contains the subscriber information and location information**
 - This database contains all the administrative information about each subscriber along with their last known location. In this way, the GSM network is able to route calls to the relevant base station for the MS.
 - When a user switches on their phone, the phone registers with the network and from this it is possible to determine which BTS it communicates with so that incoming calls can be routed appropriately. Even when the phone is not active (but switched on) it re-registers periodically to ensure that the network (HLR) is aware of its latest position.
 - There is one HLR per network, although it may be distributed across various sub-centres to for operational reasons.
 - **Visitor Location Register (VLR):**
 - This contains the IMSI (International Mobile Subscriber Identity) and the customer information for each roaming subscriber who is visiting the coverage area of a particular MSC.
 - **Equipment Identity Register (EIR):**
 - The EIR is the entity that decides whether a given mobile equipment may be allowed onto the network. Each mobile equipment has a number known as the International Mobile Equipment Identity.
 - This number, as mentioned above, is installed in the equipment and is checked by the network during registration. Dependent upon the information held in the EIR, the mobile may be

allocated one of three states - allowed onto the network, barred access, or monitored in case it creates problems.

- **Authentication Centre (AuC):**
- The AuC is a protected database that contains the secret key contained in the user's SIM card. It is used for authentication.
- **Mobile Switching Centre (MSC):**
- The MSC is the central unit in the NSS and controls the traffic among all of the BSCs.
- **Base Station Subsystem (BSS)**
- The Base Station Subsystem (BSS) provides and manages the radiotransmission paths between mobile stations and MSC.
- It also manages the radio interface between MS and all other subsystems of GSM.
- It consists of two elements:
 - 1) **Base Transceiver Station (BTS):** The BTS used in a GSM network comprises the radio transmitter receivers, and their associated antennas that transmit and receive to directly communicate with the mobiles. The BTS communicates with the mobiles.
 - 2) **Base Station Controller (BSC):** It controls a group of BTSs, and is often co-located with one of the BTSs in its group. It manages the radio resources and controls items such as handover within the group of BTSs, allocates channels.
- Base stations are carefully located by the network provider to enable complete coverage of an area. The area being covered by a base station often being referred to as a cell.
- It is important to have the different BTSs linked with the BSS and the BSSs linked back to the core network.
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- 3) **Mobile station**
- Mobile stations (MS), mobile equipment (ME) or as they are most widely known, cell or mobile phones are the section of a GSM mobile communications network that the user sees and operates.
- There are a number of elements to the cell phone, although the two main elements are the main hardware and the SIM.
- The hardware itself contains the main elements of the mobile phone including the display, case, battery, and the electronics used to generate the signal, and process the data receiver and to be transmitted.
- The mobile station, or ME also contains a number known as the International Mobile Equipment Identity (IMEI). This is installed in the phone at manufacture and "cannot" be changed. It is accessed by the network during registration to check whether the equipment has been reported as stolen.

- **The SIM or Subscriber Identity Module** contains the information that provides the identity of the user to the network. It contains a variety of information including a number known as the International Mobile Subscriber Identity (IMSI). As this is included in the SIM, and it means that by moving the SIM card from one mobile to another, the user could easily change mobiles.
- **Operation and Support Subsystem (OSS)**
- The OSS or operation support subsystem is an element within the overall GSM mobile communications network architecture that is connected to components of the NSS and the BSC. It is used to control and monitor the overall GSM network and it is also used to control the traffic load of the BSS

➤ **Applications of Mobile communication**

- It is used for personal communication and cellular phones offer voice and data connectivity with high speed.
- Transmission of news across the globe is done within a few seconds.
- Using Internet of Things (IoT), it is made possible to control various devices from a single device. Example: home automation using a mobile phone.
- It enables smart classrooms, online availability of notes, monitoring student activities etc. in the field of education.

➤ **Advantages of Mobile Communication**

- There are following advantages of mobile communication:
- **Flexibility:** Wireless communication enables the people to communicate with each other regardless of location. There is no need to be in an office or some telephone booth in order to pass and receive messages.
- **Cost effectiveness:** In wireless communication, there is no need of any physical infrastructure (Wires or cables) or maintenance practice. Hence, the cost is reduced.
- **Speed:** Improvements can also be seen in speed. The network connectivity or the accessibility was much improved in accuracy and speed.
- **Accessibility:** With the help of wireless technology easy accessibility to the remote areas is possible.
- **Constant connectivity:** Constant connectivity ensures that people can respond to emergencies relatively quickly. For example, a wireless device like mobile can ensure you a constant connectivity though you move from place to place or while you travel, whereas a wired landline can't.
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➤ Disadvantages of Mobile Communication

- Less secure due to communication through open space.
- Increased chance of interference and jamming.
- Slower transmission speed.
- Limited bandwidth and network security.
- Constant distractions and social disruption.

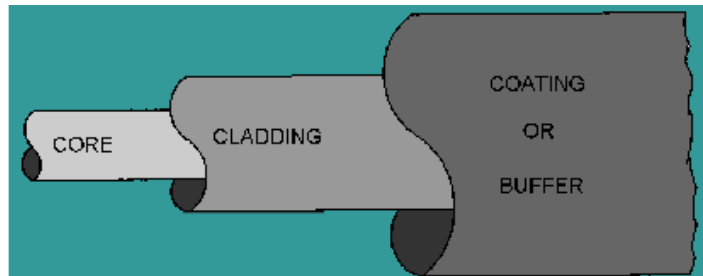
4.7 FIBER OPTIC COMMUNICATION (OPTICAL FIBER COMMUNICATION SYSTEM)

- A communication system that uses light as the carrier of information from a source to a destination through a guided fiber cable (glass or plastic) is called an optical fiber communication system.
- Light frequencies used in fiber optical systems are between 10000 to 400,000 GHz , and as a result they have higher information-carrying capacity.
- In addition to the capability of carrying a tremendous amount of information, fibers have an extremely low loss of about 0.2dB/km i.e., 0.5% of power loss over a distance of 1 km.
- Because of high information-carrying capacity and low attenuation, nowadays fibers are finding a wide application in telecommunication, local area networks, sensors, computer networks, etc.

➤ The optical fiber

Basic structure of an optical fiber

- The basic structure of an optical fiber consists of three parts; the **core**, the **cladding**, and the **coating** or **buffer**. The basic structure of an optical fiber is shown in figure below.
- The **core** is a cylindrical rod of dielectric material. Dielectric material conducts no electricity. Light propagates mainly along the core of the fiber.
- The core is generally made of glass. The **core** is described as having a radius of 'a' and a refractive index of n_1 .



- The core is surrounded by a layer of material called the **cladding**. Even though light will propagate along the fiber core without the layer of cladding material, the cladding does perform some necessary functions.
- The **cladding** layer is made of a dielectric material with an index of refraction n_2 . The index of refraction of the cladding material is less than that of the core material. The cladding is generally made of glass or plastic.
- The cladding performs the following functions:
 - ☐ Reduces loss of light from the core into the surrounding air
 - ☐ Reduces scattering loss at the surface of the core
 - ☐ Protects the fiber from absorbing surface contaminants
 - ☐ Adds mechanical strength
- For extra protection, the cladding is enclosed in an additional layer called the coating or buffer.
- The coating or buffer is a layer of plastic material used to protect an optical fiber from physical damage. The buffer is elastic in nature and prevents abrasions. The buffer also prevents the optical fiber from scattering losses caused by microbends. Microbends occur when an optical fiber is placed on a rough and distorted surface.

➤ Principle of Optical Fiber Communication

- The transmission of light in an optical fiber involves the phenomena of total internal reflections at the interface between the core and cladding.
- When light enters one end of a glass fiber under the right conditions, most of the light will propagate or move down the length of the fiber and exit from the far end.
- A small part of the light will escape through the sidewalls of the fiber, and some will also be lost due to internal absorptions. But, a portion of the light will be contained and guided to the far end. Such fiber is called a “light pipe” or “light guide”.

- Total internal reflection within the fiber wall can occur only if two conditions are satisfied.
 - 1) The refractive index n_1 of the core must always be greater than the refractive index n_2 of the cladding.
 - 2) The light must approach the wall with an angle of the incidence ϕ that is greater than the critical angle ϕ_c given by,

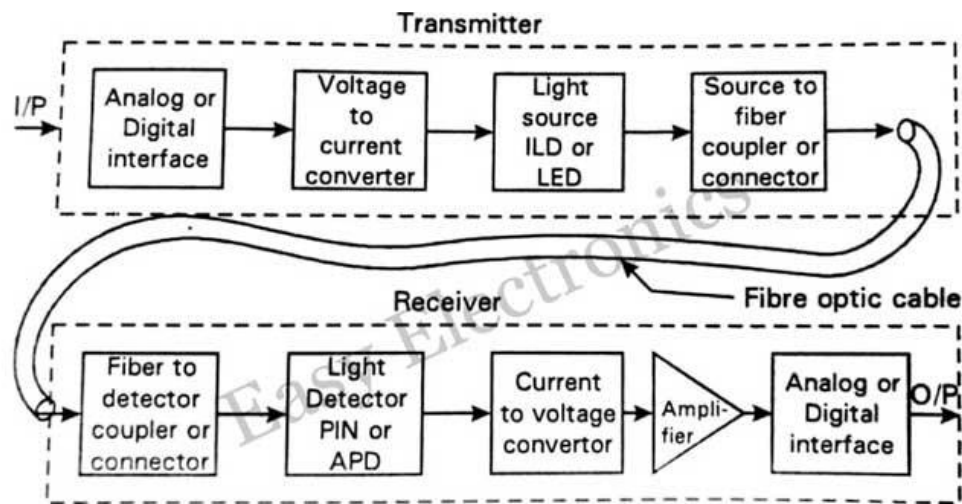
$$\sin \phi_c = n_1 / n_2$$

When $\phi = \phi_c$, then by snell's law,

$$n_1 \sin \phi_c = n_2 \sin 90^\circ$$

➤ **Block diagram of Optical Fiber Communication system**

- Fiber optic communication link is the transmission of information by the propagation of the optical signal through optical fibers over a required distance. This involves deriving an optical signal from an electrical signal at the transmission end and conversion of the optical signal back to an electrical signal at the receiving end.
- The important components of fiber optic communication links are:
 1. Transmitter
 2. Fiber optic cable
 3. Receiver



Block diagram of optical fiber communication system

<https://easyelectronics.co.in>

Transmitter

The transmitter consists of an analog or digital interface, a voltage-to-current converter, a light source, and a source-to-fiber connector or coupler.

Fibre Optic Cable

The fiber optic cable is either an ultra-pure glass or plastic cable.

Receiver

The receiver includes a fiber-to-detector coupler, a light detector, a current-to-voltage converter followed by an amplifier, and an analog or digital interface.

Working of Optical fiber communication system

- The information source provides an electrical signal to a transmitter consisting of an electrical stage that drives the optical source.
- The optical source (light source) in the transmitter converts the electrical signals to an optical signal (i.e; it gives modulation of the light wave).
- The two types of optical sources are light-emitting diodes (LEDs) and laser diodes. The optical source launches the optical signal into the fiber through the optical connectors or couplers.
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- The optical signal will be guided along the fiber by the phenomenon of total internal reflection. The optical wave becomes progressively weakened and distorted because of scattering, absorption, and dispersion mechanisms in the fiber waveguides.
- The receiver converts the optical signal exiting the fiber back into an electrical signal. The receiver consists of two parts, the optical detector and the signal-conditioning circuits.
- An optical detector detects the optical signal (demodulation of the optical carrier) and converts the optical signal back to the electrical form. An optical detector can be either a semiconductor positive-intrinsic-negative (PIN) diode or an avalanche photodiode (APD).
- The signal-conditioning circuit conditions the detector output so that the receiver output matches the original input to the transmitter.

➤ **Advantages of optical fiber communication**

The primary advantages of optical fiber communication compared to metallic cable (electrical) communication are as under:

- 1. Extremely Wide (Large) Bandwidth:** The bandwidth available with single glass fiber is more than 100 GHz.
- 2. Immunity to Electrostatic Interference:** As optical fibers are made of either glass or plastic (non conductors of electricity) external electrical noise and lightning do not affect the energy in a fiber cable.
- 3. Immunity to interference and crosstalks:** Fiber systems are immune to cross-talk between cables caused by magnetic induction. Whereas, in metallic cables, cross-talk results from the electromagnetic coupling between two adjacent wires.
- 4. Lighter Weight and Smaller Size:** Fibers are very small in size. Reduction in size results in a reduction of weight also.
- 5. Lower Cost:** The material used in fibers is silica glass or silicon dioxide which is one of the most abundant materials on earth, resulting in lower cost.
- 6. Security:** Fiber cables are more secure than metallic cables. Due to its immunity to electromagnetic coupling and radiation, optical fiber can be used in the most secure environments.
- 7. Greater Safety due to electrical isolation:** In many wired systems, (metallic cables) the potential hazard of short circuits require precautionary designs. Whereas, the dielectric nature of optical fibers eliminates the spark hazard.

8. Corrosion: Fiber cables are more resistant to environmental extremes. They operate over large temperature variations than their metallic counterparts and are less affected by corrosive liquids and gases.

9. Longer Life Span and Ease of Maintenance: A longer life span of 20 to 30 years is predicted for fiber optic cables as compared to 12 to 15 years for conventional cables.

10. Low transmission loss: Fibers have an extremely low loss of about 0.2dB/km i.e., 0.5% of power loss over a distance of 1 km.

11. Ruggedness and flexibility: Optical fibers are manufactured with high tensile strengths and so they can be bent or twisted without damage.

➤ **Application of optical fiber communication**

The application of optical fibers in various fields is given below:

1. Optical Fibres used in the Medical industry: Because of its extremely thin and flexible nature, it is used in various instruments to view internal body parts by inserting them into hollow spaces in the body. It is used as a laser during surgeries, endoscopy, microscopy, and biomedical research.

2. Optical Fibres Used in Communication: In the communication system, telecommunication has major uses of optical fiber cables for transmitting and receiving purposes. It is used in various networking fields and even increases the speed and accuracy of the transmission data. Compared to copper wires, fiber optics cables are lighter, more flexible, and carry more data.

3. Optical Fibres used for Defense Purposes: Fiber optics are used for data transmission in high-level data security fields of military and aerospace applications. These are used in wirings in aircraft, hydrophones for SONARs, and seismic applications.

4. Optical Fibres are used in Industries: These fibers are used for imaging in hard-to-reach places such as they are used for safety measures and lighting purposes in automobiles both in the interior and exterior. They transmit information at lightning speed and are used in airbags and traction control. They are also used for research and testing purposes in industries.

5. Optical Fibres used for Broadcasting: These cables are used to transmit high-definition television signals which have greater bandwidth and speed. Optical Fibre is cheaper compared to the same quantity of copper wires. Broadcasting companies use optical fibers for wiring HDTV, CATV, video-on-demand, and many other applications.

6. Uses of Optical Fibre for Lightening and Decorations: Optical fibers also gives an attractive, economical, and easy way to illuminate the area and that is why it is widely used in decorations and Christmas trees.

7. Optical Fibres used in Mechanical Inspections: On-site inspection engineers use optical fibers to detect damages and faults which are hard-to-reach places. Even plumbers use optical fibers for the inspection of pipes.

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